Methods In Calculus

Chapter Overview

Improper integrals

Replace an infinite limit or an undefined endpoint with a limit.

Check convergence before evaluating.

Further techniques

Find the mean value of a function.

Differentiate and integrate inverse trigonometric functions.

Use partial fractions in integration.

1. Why does direct substitution give an impossible result for $\int_{-1}^{1} x^{-2}\,dx$?

1

$$[-x^{-1}]_{-1}^{1} = -2$$

This is invalid: the integrand is undefined at zero. Split the integral there. Each positive part diverges, so the improper integral diverges.

2. Evaluate $\int_{1}^{\infty} x^{-2}\,dx$.

2

$$\lim_{b\to\infty}[-x^{-1}]_{1}^{b} = \lim_{b\to\infty}(1 - 1/b) = 1$$

Undefined endpoints

Evaluate $\int_{0}^{1} x^{-1/2}\,dx$.

Endpoint limit

$$\lim_{t\to 0^{+}}[2\sqrt{x}]_{t}^{1} = 2$$

Infinite intervals

An integral can converge even though its interval is unbounded. For example, the normal density has total area 1:

$$\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-z^{2}/2}\,dz = 1$$

Always check that each required limit exists.

Examples

Solution 5

Compare $\int_1^\infty x^{-2}\,dx$ and $\int_1^\infty x^{-1}\,dx$.

Compare

$$\int_1^b x^{-2}\,dx = 1 - 1/b \to 1$$

$$\int_1^b x^{-1}\,dx = \ln b \to \infty$$

A function tending to zero is not sufficient for its integral to converge.

Similarly, the harmonic series $\sum_{r=1}^\infty 1/r$ diverges despite its terms tending to zero.

When f(x) not defined for some value

Solution 6

1. Evaluate or establish divergence: $\int_0^1 x^{-2}\,dx$.

1

$$\lim_{t\to 0^+}\left[-x^{-1}\right]_t^1 = \lim_{t\to 0^+}(1/t - 1) = \infty$$

The integral diverges.

2. Evaluate $\int_0^2 x/\sqrt{4-x^2}\,dx$.

2

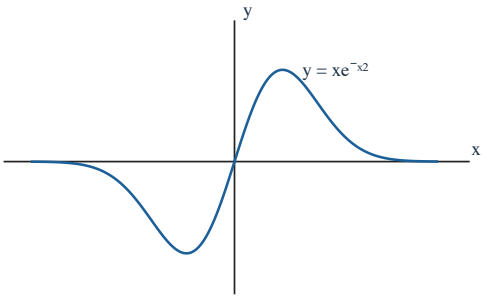
$$\lim_{b\to 2^-}\left[-\sqrt{4-x^2}\right]_0^b = 2$$

An improper integral with two infinite limits

Solution 7

Evaluate $\int_{-\infty}^{\infty} xe^{-x^2}\,dx$.

Split at zero; both integrals must converge separately.



Integral

$$\int xe^{-x^2}\,dx = -\frac{1}{2}e^{-x^2} + C$$

$$\int_{-\infty}^{0} xe^{-x^2}\,dx = -\frac{1}{2}$$

$$\int_{0}^{\infty} xe^{-x^2}\,dx = \frac{1}{2}$$

$$\int_{-\infty}^{\infty} xe^{-x^2}\,dx = 0$$

Exercise 3A - Core Pure Year 2 Pages 56-58

Solution 8

5a. Find $\int (7-3x)^{-2}\,dx$.

b. Hence evaluate the integral from $-\infty$ to 2.

a

$$\int (7-3x)^{-2}\,dx = \frac{1}{3(7-3x)} + C$$

b

$$\lim_{t\to-\infty}\left[\frac{1}{3(7-3x)}\right]_t^2 = \frac{1}{3}$$

6. Find $\int x^2 e^{x^3}\,dx$, then evaluate $\int_{-\infty}^{1} x^2 e^{x^3}\,dx$.

Integrate

$$u = x^3, \quad du = 3x^2\,dx$$

$$\int x^2 e^{x^3}\,dx = \frac{1}{3}e^{x^3} + C$$

Limit

$$\lim_{t\to-\infty} \frac{1}{3}(e - e^{t^3}) = \frac{e}{3}$$

7. Find $\int \ln x/x\,dx$. Does $\int_1^{\infty} \ln x/x\,dx$ converge?

Integrate

$$\int \frac{\ln x}{x}\,dx = \frac{1}{2}(\ln x)^2 + C$$

Convergence

$$\lim_{b\to\infty} \frac{1}{2}(\ln b)^2 = \infty$$

The integral diverges.

8. Integrate $(\ln x)^2$. Hence evaluate the integral over $(0, 1]$ and discuss $(0, \infty)$.

Integrate

Integrating by parts twice:

$$\int (\ln x)^2 dx = x((\ln x)^2 - 2\ln x + 2) + C$$

Limits

As $t \to 0^+$, both $t \ln t$ and $t(\ln t)^2$ tend to zero.

$$\int_0^1 (\ln x)^2 dx = 2$$

The integral over $(0, \infty)$ diverges at infinity.

9. Evaluate $\int_0^2 6x(4 - x^2)^{-1/3}\, dx$.

9

$$\lim_{b \to 2^-} \left[-\frac{9}{2}(4 - x^2)^{2/3} \right]_0^b$$

$$= \frac{9}{2} 4^{2/3} = 9\sqrt[3]{2}$$

10. Evaluate

$$\int_{-2}^{2} \frac{\sqrt{2 - x} - 3\sqrt{2 + x}}{\sqrt{4 - x^2}}\, dx$$

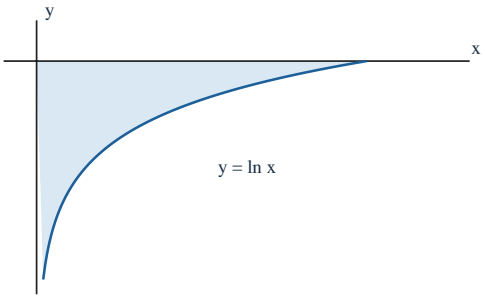
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$$= \int_{-2}^{2} \left(\frac{1}{\sqrt{2 + x}} - \frac{3}{\sqrt{2 - x}} \right) dx$$

Take the endpoint limits separately; both converge.

$$= 4 - 12 = -8$$

11. Find the finite area bounded by $y = \ln x$, the x-axis and the y-axis. You may use $x \ln x \to 0$ as $x \to 0^+$.



11

The curve is below the axis on $0 < x < 1$.

$$A = -\lim_{t \to 0^+} \int_t^1 \ln x \, dx$$

$$= -\lim_{t \to 0^+} [x \ln x - x]_t^1 = 1$$

12a. Explain why $\int_0^{\pi/2} \tan x \, dx$ is improper.

b. Determine whether it converges.

a

$\tan x$ is undefined at $x = \pi/2$.

b

$$\lim_{b \to (\pi/2)^-} [-\ln(\cos x)]_0^b = \infty$$

The integral diverges.

The Mean Value of a Function

Solution 15

Mean value of a function

The discrete mean is $(y_1 + \cdots + y_n)/n$. For equally spaced samples, a limiting average gives:

Continuous mean

$$\overline{f} = \frac{1}{b-a}\int_a^b f(x)\,dx$$

The integral is the signed area; divide by the interval width.

A constant function has that same constant as its mean.

Examples

Solution 16

1. Find the mean of $4/\sqrt{2+3x}$ on $[2, 6]$.

1

$$\overline{f} = \frac{1}{4}\left[\frac{8}{3}\sqrt{2+3x}\right]_2^6$$
$$= \frac{4}{3}(\sqrt{5} - \sqrt{2})$$

2. For $f(x) = 4/(1+e^x)$, find the means of f, $f+4$, and $-f$ on $[\ln 2, \ln 6]$.

2: integrate

Let $u = e^x$.

$$\int_2^6 \frac{4}{u(1+u)}du = 4[\ln u - \ln(1+u)]_2^6$$
$$= 4\ln(9/7)$$

2: means

$$M = \frac{4\ln(9/7)}{\ln 3}$$

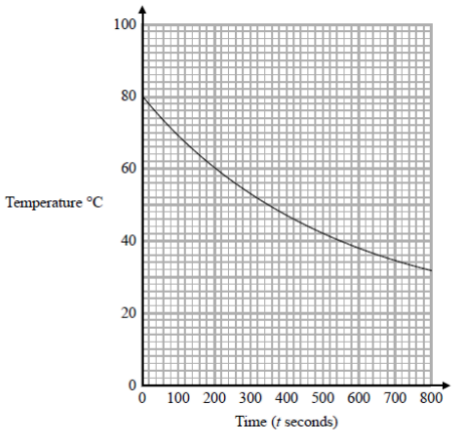
The other means are $M + 4$ and $-M$.

Just for your interest...

Solution 17

Why does the gradient joining the first and last points give the average rate of change?

Apply this to water cooling from $80°\text{C}$ between $t = 0$ and $t = 800$.



Average gradient

$$\overline{f'} = \frac{1}{b-a}\int_a^b f'(t)dt$$

$$= \frac{f(b)-f(a)}{b-a}$$

This is exactly the secant gradient.

Reading $T(800) \approx 32°\text{C}$, the average rate of decrease is

$(80-32)/800 = 0.06°\text{C}/s$.

Exercise 3B - Core Pure Year 2 Pages 61-62

Solution 18

4. Find the exact mean of

$$f(x) = \frac{\sin x \cos x}{\cos 2x + 2}, \qquad 0 \le x \le \pi/2$$

4

$$\int f(x)dx = -\frac{1}{4}\ln(\cos 2x + 2) + C$$

$$\overline{f} = \frac{2}{\pi}\left[-\frac{1}{4}\ln(\cos 2x + 2)\right]_0^{\pi/2}$$

$$= \frac{\ln 3}{2\pi}$$

6. Find the exact mean of $f(x) = x \sin 2x$ on $[0, \pi/3]$.

Solution 19

6

Integrate by parts:

$$\int x \sin 2x \, dx = -\frac{x}{2}\cos 2x + \frac{1}{4}\sin 2x + C$$

$$\overline{f} = \frac{3}{\pi}\left(\frac{\pi}{12} + \frac{\sqrt{3}}{8}\right) = \frac{1}{4} + \frac{3\sqrt{3}}{8\pi}$$

7. Let $f(x) = 5x/((2x-1)(x+2))$.
   a. Show its mean on $[1, 5]$ is $\frac{1}{4}\ln(49/3)$.
   b. Find the mean of $f(x) + \ln k$, $k > 0$, in the form $p \ln q$.

Solution 20

a

$$f(x) = \frac{1}{2x-1} + \frac{2}{x+2}$$

$$\overline{f} = \frac{1}{4}[\tfrac{1}{2}\ln(2x-1) + 2\ln(x+2)]_1^5$$

$$= \frac{1}{4}\ln\frac{49}{3}$$

b

$$\overline{f} + \ln k = \frac{1}{4}\ln\frac{49k^4}{3}$$

Exercise 3B - Core Pure Year 2 Pages 61-62

Solution 21

8. Let $f(x) = x(x^2 - 4)^4$.

a. Show its mean on $[0, 2]$ is $256/5$.

b. Hence find the mean of $-2f(x)$.

a

$$\overline{f} = \frac{1}{2}\left[\frac{(x^2-4)^5}{10}\right]_0^2 = \frac{256}{5}$$

b

$$\overline{-2f} = -2\overline{f} = -\frac{512}{5}$$

Exercise 3B - Core Pure Year 2 Pages 61-62

Solution 22

11. Find the exact mean of $f(x) = 1/\sqrt{2-x}$ over $[0, 2]$.

11

$$\overline{f} = \frac{1}{2}\lim_{b\to 2^-}\left[-2\sqrt{2-x}\right]_0^b = \sqrt{2}$$

The endpoint is undefined but the integral converges.

Exercise 3B - Core Pure Year 2 Pages 61-62 — Solution 23

13. Let $f(x) = \cos x/(2+\sin x)^2$.

a. Integrate f.

b. Find its mean over $[0, 5\pi/3]$.

c. Find the mean of $f(x) + 3x$ over that interval.

a

$$\int f(x)dx = -\frac{1}{2+\sin x} + C$$

b

$$M = \frac{3}{5\pi}\left[-\frac{1}{2+\sin x}\right]_0^{5\pi/3}$$
$$= -\frac{3}{130\pi}(3+4\sqrt{3})$$

c

$$\overline{f+3x} = M + \frac{5\pi}{2}$$

Differentiating inverse trigonometric functions — Solution 24

Show that $d(\arcsin x)/dx = 1/\sqrt{1-x^2}$.

Derivation

Let $x = \sin y$, where $-\pi/2 < y < \pi/2$.

$$\frac{dx}{dy} = \cos y = \sqrt{1-x^2}$$
$$\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}\arccos x = -\frac{1}{\sqrt{1-x^2}}$$
$$\frac{d}{dx}\arctan x = \frac{1}{1+x^2}$$

Differentiate $y = \arcsin(x^2)$.

Chain rule

$$\frac{dy}{dx} = \frac{2x}{\sqrt{1-x^4}}$$

Test Your Understanding

Solution 25

1. Differentiate $y = \arctan((1-x)/(1+x))$.

2. Differentiate $y = \operatorname{arcsec}(2x)$, for $x > 1/2$.

1

$$u = \frac{1-x}{1+x}, \quad u' = \frac{-2}{(1+x)^2}$$

$$y' = \frac{-2}{(1+x)^2 + (1-x)^2} = -\frac{1}{1+x^2}$$

2

$$\sec y = 2x \Rightarrow \sec y \tan y \, y' = 2$$

$$y' = \frac{1}{x\sqrt{4x^2-1}}$$

For the usual full principal domain, replace x in the denominator by $|x|$.

Exercise 3C - Core Pure Year 2 Pages 64-65

Solution 26

3. Differentiate

$$y = \frac{1+\arctan x}{1-\arctan x}$$

3

Use the quotient rule:

$$y' = \frac{(1-\arctan x) + (1+\arctan x)}{(1+x^2)(1-\arctan x)^2}$$

$$= \frac{2}{(1+x^2)(1-\arctan x)^2}$$

4. Prove that $\arccos x + \arcsin x = \pi/2$ for $-1 \le x \le 1$.

4

$$\frac{d}{dx}(\arccos x + \arcsin x) = 0 \qquad (-1 < x < 1)$$

The sum is constant. At zero it is $\pi/2 + 0$. Continuity includes the two endpoints.

6. Given $\tan y = x \arctan x$, find dy/dx in terms of x.

6

$$\sec^2 y\, y' = \arctan x + \frac{x}{1+x^2}$$

$$y' = \frac{\arctan x + x/(1+x^2)}{1 + x^2(\arctan x)^2}$$

7. For $y = \arcsin x$, prove $(1-x^2)y'' - xy' = 0$.

7

$$y' = \frac{1}{\sqrt{1-x^2}}, \quad y'' = \frac{x}{(1-x^2)^{3/2}}$$

$$(1-x^2)y'' = \frac{x}{\sqrt{1-x^2}} = xy'$$

Exercise 3C - Core Pure Year 2 Pages 64-65 — Solution 29

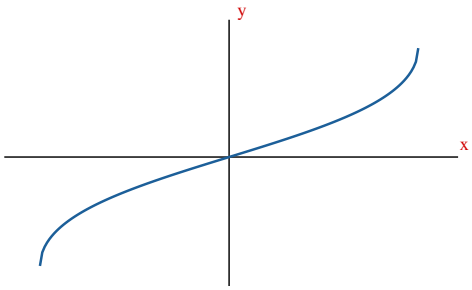
a. $y = \arcsin(\arcsin x)$ b. $y = \arccos(\arccos x)$ c. $y = \arctan(\arctan x)$

a

Odd and increasing; passes through the origin.

Domain $[-\sin 1, \sin 1]$; range $[-\pi/2, \pi/2]$. Vertical tangents at both endpoints.

a graph

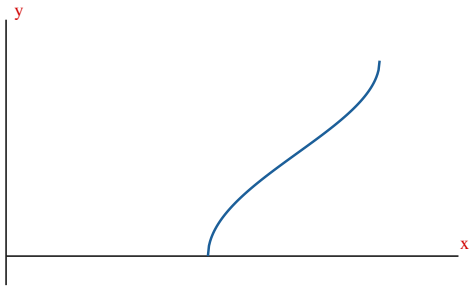


b

Increasing from $(\cos 1, 0)$ to $(1, \pi/2)$.

Domain $[\cos 1, 1]$; range $[0, \pi/2]$. Vertical tangents at both endpoints.

b graph



c

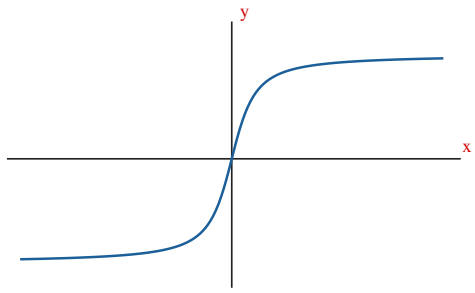
Odd and increasing through the origin.

Domain $\mathbb{R}$.

Range: $-a < y < a$, where $a = \arctan(\pi/2)$.

Horizontal asymptotes: $y = \pm a$.

c graph



Integrating with inverse trigonometric functions — Solution 30

Use a substitution to prove

$$\int \frac{1}{1+x^2}\,dx = \arctan x + C$$

Substitution

$x = \tan\theta, \quad dx = \sec^2\theta\,d\theta$

Integrate

$$\int \frac{\sec^2\theta}{1+\tan^2\theta}\,d\theta = \int 1\,d\theta$$

$$= \theta + C = \arctan x + C$$

Integrals leading to inverse sine

Solution 31

For $a > 0$ and $|x| < a$, show

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \arcsin(x/a) + C$$

Proof

$u = x/a, \quad dx = a\,du$

$$\int \frac{a}{a\sqrt{1-u^2}} du = \arcsin u + C$$

$$= \arcsin(x/a) + C$$

Example

Solution 32

Standard results

$$\int \frac{dx}{\sqrt{a^2 - x^2}} = \arcsin(x/a) + C$$

$$\int \frac{dx}{a^2 + x^2} = \frac{1}{a} \arctan(x/a) + C$$

Here $a > 0$; the first formula requires $|x| < a$.

Find $\int 4/(5 + x^2)\, dx$.

Example

$$\frac{4}{\sqrt{5}} \arctan \frac{x}{\sqrt{5}} + C$$

Inverse trigonometric integrals: practice

Solution 33

1. Find $\int 1/(25+9x^2)\,dx$.

2. Evaluate
$$\int_{-\sqrt{3}/4}^{\sqrt{3}/4} \frac{dx}{\sqrt{3-4x^2}}$$

1
$$\frac{1}{9}\int \frac{dx}{(5/3)^2+x^2}$$
$$= \frac{1}{15}\arctan(3x/5)+C$$

2
$$\frac{1}{2}\left[\arcsin(2x/\sqrt{3})\right]_{-\sqrt{3}/4}^{\sqrt{3}/4} = \frac{\pi}{6}$$

Final example

Solution 34

Find
$$\int \frac{x+4}{\sqrt{1-4x^2}}dx$$

First term
$$\int \frac{x\,dx}{\sqrt{1-4x^2}} = -\frac{1}{4}\sqrt{1-4x^2}+C$$

Second term
$$4\int \frac{dx}{\sqrt{1-4x^2}} = 2\arcsin(2x)+C$$

Combined
$$-\frac{1}{4}\sqrt{1-4x^2}+2\arcsin(2x)+C$$

4. Find constants A, B such that

$$\int \frac{dx}{4+3x^2} = A\arctan(Bx) + C$$

4

$$\frac{1}{3}\int \frac{dx}{(2/\sqrt{3})^2 + x^2} = \frac{1}{2\sqrt{3}}\arctan\frac{\sqrt{3}x}{2} + C$$

$$A = \frac{\sqrt{3}}{6}, \quad B = \frac{\sqrt{3}}{2}$$

5. Find constants P, Q such that

$$\int \frac{dx}{\sqrt{3-4x^2}} = P\arcsin(Qx) + C$$

5

$$\frac{1}{2}\int \frac{dx}{\sqrt{(\sqrt{3}/2)^2 - x^2}} = \frac{1}{2}\arcsin\frac{2x}{\sqrt{3}} + C$$

$$P = \frac{1}{2}, \quad Q = \frac{2}{\sqrt{3}}$$

7. Evaluate $\int_{\sqrt{2}}^{\sqrt{3}} (4-x^2)^{-1/2}\,dx$.

7

$$\left[\arcsin(x/2)\right]_{\sqrt{2}}^{\sqrt{3}} = \frac{\pi}{3} - \frac{\pi}{4} = \frac{\pi}{12}$$

8. Find $\int (2+3x)/(1+3x^2)\,dx$.

8

$$2\int \frac{dx}{1+3x^2} + \frac{1}{2}\int \frac{6x\,dx}{1+3x^2}$$
$$= \frac{2}{\sqrt{3}}\arctan(\sqrt{3}x) + \frac{1}{2}\ln(1+3x^2) + C$$

9. Integrate $(2x-1)/\sqrt{2-x^2}$ in the form
$A\arcsin(x/\sqrt{2}) + B\sqrt{2-x^2} + C$.

9

$$\int \frac{2x\,dx}{\sqrt{2-x^2}} = -2\sqrt{2-x^2} + C$$
$$\int \frac{-dx}{\sqrt{2-x^2}} = -\arcsin(x/\sqrt{2}) + C$$
$$A = -1, \quad B = -2$$

Exercise 3D - Core Pure Year 2 Pages 66-67 — Solution 39

11. Find $\int (4x-1)/\sqrt{6-5x^2}\,dx$.

First term

$$\int \frac{4x\,dx}{\sqrt{6-5x^2}} = -\frac{4}{5}\sqrt{6-5x^2} + C$$

Second term

$$\int \frac{-dx}{\sqrt{6-5x^2}} = -\frac{1}{\sqrt{5}} \arcsin\left(\sqrt{\frac{5}{6}}x\right) + C$$

Combined

$$-\frac{4}{5}\sqrt{6-5x^2} - \frac{1}{\sqrt{5}} \arcsin\left(\sqrt{\frac{5}{6}}x\right) + C$$

Challenge 3D - Core Pure Year 2 Pages 66-67 — Solution 40

Challenge. For $x > 1$, use $x = \sec\theta$ to integrate:

a. $1/(x\sqrt{x^2-1})$

b. $\sqrt{x^2-1}/x$.

Substitute

$$dx = \sec\theta\tan\theta\,d\theta, \quad \sqrt{x^2-1} = \tan\theta$$

a

$$\int \frac{dx}{x\sqrt{x^2-1}} = \int d\theta = \operatorname{arcsec} x + C$$

b

$$\int \frac{\sqrt{x^2-1}}{x}dx = \int \tan^2\theta\,d\theta$$

$$= \tan\theta - \theta + C = \sqrt{x^2-1} - \operatorname{arcsec} x + C$$

Solving using partial fractions

Solution 41

Use partial fractions to prove, for $a > 0$,

$$\int \frac{dx}{a^2 - x^2} = \frac{1}{2a} \ln \left| \frac{a+x}{a-x} \right| + C$$

Split

$$\frac{1}{a^2 - x^2} = \frac{1}{2a} \left(\frac{1}{a+x} + \frac{1}{a-x} \right)$$

Integrate

$$\int \frac{dx}{a^2 - x^2} = \frac{1}{2a} (\ln |a+x| - \ln |a-x|) + C$$
$$= \frac{1}{2a} \ln \left| \frac{a+x}{a-x} \right| + C$$

Partial Fractions involving Quadratic Factors

Solution 42

Quadratic factors

Use a general linear numerator over an irreducible quadratic:

$$\frac{1}{x(x^2+1)} = \frac{A}{x} + \frac{Bx + C}{x^2 + 1}$$

Find $\int (1+x)/(x^3 + 9x)\, dx$.

Partial fractions

$$\frac{1+x}{x(x^2+9)} = \frac{1}{9x} + \frac{1 - x/9}{x^2 + 9}$$

Integral

$$\frac{1}{9} \ln |x| - \frac{1}{18} \ln(x^2+9) + \frac{1}{3} \arctan(x/3) + C$$
$$= \frac{1}{18} \ln \frac{x^2}{x^2+9} + \frac{1}{3} \arctan(x/3) + C$$

Test Your Understanding — Solution 43

a. Express $(x^4+x)/(x^4+5x^2+6)$ in partial fractions.

b. Hence integrate it.

Equal numerator and denominator degrees require a constant quotient.

a

$$\frac{x^4+x}{(x^2+2)(x^2+3)} = 1 + \frac{x+4}{x^2+2} - \frac{x+9}{x^2+3}$$

b

$$x + \frac{1}{2}\ln\frac{x^2+2}{x^2+3} + 2\sqrt{2}\arctan\frac{x}{\sqrt{2}} - 3\sqrt{3}\arctan\frac{x}{\sqrt{3}} + C$$

Exercise 3E - Core Pure Year 2 Pages 72-73 — Solution 44

2. Find $\int (x^2-3x)/((x^2+6)(x+2))\,dx$ in the form $A\ln|x+2| + B\arctan(x/\sqrt{6}) + C$.

Split

$$\frac{x^2-3x}{(x^2+6)(x+2)} = \frac{1}{x+2} - \frac{3}{x^2+6}$$

Integrate

$$\ln|x+2| - \frac{3}{\sqrt{6}}\arctan\frac{x}{\sqrt{6}} + C$$

$$A = 1, \quad B = -\frac{3}{\sqrt{6}}$$

Exercise 3E - Core Pure Year 2 Pages 72-73 Solution 45

3. Let $f(x) = x^4 - x^3 - 4x^2 - 2x - 12$.

a. Given $x + 2$ is a factor, fully factorise f.

b. Integrate $(x^3 - 20x^2 + 4x - 24)/f(x)$.

a

$$f(x) = (x + 2)(x - 3)(x^2 + 2)$$

b

$$\frac{x^3 - 20x^2 + 4x - 24}{f(x)} = \frac{4}{x + 2} - \frac{3}{x - 3} - \frac{2}{x^2 + 2}$$

$$\int = \ln \frac{|x + 2|^4}{|x - 3|^3} - \sqrt{2} \arctan \frac{x}{\sqrt{2}} + C$$

Exercise 3E - Core Pure Year 2 Pages 72-73 Solution 46

4. Find $\int (2 - x)/(x^3 + 4x)\, dx$ in the form $A \ln(x^2/(x^2 + 4)) + B \arctan(x/2) + C$.

Split

$$\frac{2 - x}{x(x^2 + 4)} = \frac{1}{2x} - \frac{x/2 + 1}{x^2 + 4}$$

Integrate

$$\frac{1}{2} \ln |x| - \frac{1}{4} \ln(x^2 + 4) - \frac{1}{2} \arctan(x/2) + C$$

$$A = \frac{1}{4}, \quad B = -\frac{1}{2}$$

6. Integrate $(x^3 + 9x^2 + x + 1)/(x^4 - 1)$, giving a logarithm of a quotient plus $D \arctan x$.

Split

$$\frac{x^3 + 9x^2 + x + 1}{x^4 - 1} = \frac{3}{x - 1} - \frac{2}{x + 1} + \frac{4}{x^2 + 1}$$

Integrate

$$\ln \frac{|x - 1|^3}{|x + 1|^2} + 4 \arctan x + C$$

$$A = 3, \quad B = 2, \quad D = 4$$

7. Let $f(x) = x^3 - 4x^2 + 6x - 24$.

a. Given $f(4) = 0$, factorise f.

b. Express $(2x^2 - 3x + 24)/f(x)$ in partial fractions.

c. Hence integrate it.

a

$$f(x) = (x - 4)(x^2 + 6)$$

b

$$\frac{2x^2 - 3x + 24}{f(x)} = \frac{2}{x - 4} - \frac{3}{x^2 + 6}$$

c

$$2 \ln |x - 4| - \frac{3}{\sqrt{6}} \arctan \frac{x}{\sqrt{6}} + C$$

Exercise 3E - Core Pure Year 2 Pages 72-73 — Solution 49

8. Let $f(x) = 1/((x-2)(2x-1))$.
   a. Integrate f.
   b. Show $\int_{1/2}^{2} f(x)\,dx$ diverges.

a

$$f(x) = \frac{1}{3(x-2)} - \frac{2}{3(2x-1)}$$

$$F(x) = \frac{1}{3}\ln\left|\frac{x-2}{2x-1}\right| + C$$

b

Split at 1. As $b \to 2^-$, $F(b) \to -\infty$, so $\int_1^2 f(x)dx$ diverges.

The full integral therefore diverges (to $-\infty$).

Exercise 3E - Core Pure Year 2 Pages 72-73 — Solution 50

9. Express $(x^4+5x^2+2x)/(x^4+10x^2+24)$ in partial fractions, then integrate it.

a

$$\frac{x^4+5x^2+2x}{(x^2+4)(x^2+6)} = 1 + \frac{x-2}{x^2+4} - \frac{x+3}{x^2+6}$$

b

$$x + \frac{1}{2}\ln\frac{x^2+4}{x^2+6} - \arctan(x/2) - \frac{3}{\sqrt{6}}\arctan(x/\sqrt{6}) + C$$

Exercise 3E - Core Pure Year 2 Pages 72-73 Solution 51

10. Use partial fractions to integrate $(x^2 + 4x + 10)/(x^3 + 5x)$, for $x > 0$.

Split

$$\frac{x^2 + 4x + 10}{x(x^2 + 5)} = \frac{2}{x} + \frac{4 - x}{x^2 + 5}$$

Integrate

$$2\ln x - \frac{1}{2}\ln(x^2 + 5) + \frac{4}{\sqrt{5}}\arctan\frac{x}{\sqrt{5}} + C$$

Exercise 3E - Core Pure Year 2 Pages 72-73 Solution 52

11. Evaluate $\int_0^2 2/((x + 1)(x^2 + 1))\,dx$.

Check carefully whether the claimed value $(\pi + 2\ln 2)/4$ matches this upper limit.

Integrate

$$\frac{2}{(x + 1)(x^2 + 1)} = \frac{1}{x + 1} + \frac{1 - x}{x^2 + 1}$$

$$F(x) = \ln(x + 1) + \arctan x - \tfrac{1}{2}\ln(x^2 + 1)$$

Evaluate

$$\int_0^2 = \ln 3 + \arctan 2 - \tfrac{1}{2}\ln 5$$

The claimed value instead corresponds to upper limit 1.

12. Express $(x^4+1)/(x(x^2+2)^2)$ in partial fractions, then integrate it.

a

$$\frac{x^4+1}{x(x^2+2)^2} = \frac{1}{4x} + \frac{3x}{4(x^2+2)} - \frac{5x}{2(x^2+2)^2}$$

b

For the last term, substitute $u = x^2+2$.

$$\int = \frac{1}{4}\ln|x| + \frac{3}{8}\ln(x^2+2) + \frac{5}{4(x^2+2)} + C$$